



Q1 Prime Number

10 marks

A mathematical analysis system needs to verify whether a given number is prime. A prime number is a number greater than 1 that has no divisors other than 1 and itself.

Write a Python program that:

- Accepts an integer input from the user
- Checks whether the number is prime using appropriate control flow constructs
- Displays a suitable message indicating whether the number is prime or not

Ensure proper use of loops and conditional statements.

Your Code

```
1 n=int(input("Enter the number:\n"))
2 if n<=1:
3     print("Not a Prime Number")
4 else:
5     for i in range (2,n):
6         if (n%i==0):
7             print("Not a Prime Number")
8             break
9     else:
10        print("Prime Number")
```

Input

7

Output

```
Enter the number:
Prime Number
```

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Q2 Fibonacci series

10 marks

A numerical computation module is designed to generate a sequence where each number is the sum of the two preceding numbers. This sequence is known as the Fibonacci series.

Write a Python program that:

- Accepts a number n from the user
- Generates and displays the first n terms of the Fibonacci sequence
- Uses looping constructs to compute the sequence

Ensure the program handles cases where n is less than or equal to zero.

Your Code

```
1 n=int(input("Enter the number:\n"))
2 i=0
3 a=0
4 b=1
5 for i in range (1,n+1):
6     print(a)
7     c=a+b
8     a=b
9     b=c
10    i=i+1
```

Input

10

Output

```
Enter the number:
0
1
1
2
3
```

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Q3 String

10 marks

A text processing system receives user input in the form of a string that may contain alphabets, digits, and special characters.

Write a Python program that:

- Accepts a string input from the user
- Counts the number of vowels, consonants, digits, and special characters present in the string
- Displays the count for each category

Use appropriate string operations and control flow statements.

Your Code

```
1 text = input("Enter a string: ")
2 vowels = 0
3 consonants = 0
4 digits = 0
5 special_characters = 0
6 for ch in text:
7     if ch.isalpha():
8         if ch.lower() in "aeiou":
9             vowels += 1
10        else:
11            consonants += 1
12    elif ch.isdigit():
13        digits += 1
14    else:
15        special_characters += 1
16 print("\nCharacter Count:")
17 print("Vowels:", vowels)
18 print("Consonants:", consonants)
19 print("Digits:", digits)
20 print("Special Characters:", special_characters)
```

Input

#Riyaz987

Output

```
Enter a string:
Character Count:
Vowels: 2
Consonants: 3
Digits: 3
Special Characters: 1
```

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Q4 ODD or EVEN

10 marks

A program is designed to check whether a number entered by the user is even or odd. However, the program contains syntax and logical errors.

Analyze the given code and perform the following:

- Identify all errors present in the program
- Rewrite the corrected program
- Briefly justify each correction

```
num = input("Enter number: ")  
if num % 2 == 0  
print("Even")  
else  
print("Odd")
```

Your Code

```
1 num=int(input("Enter number:\n"))  
2 if num%2==0:  
3     print("Even")  
4 else:  
5     print("Odd")
```

Input

49

Output

```
Enter number:  
Odd
```

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Q5 Print numbers from 1 to 10

10 marks

A program is intended to display numbers from 1 to 10 using a loop. However, some parts of the code are missing.

Complete the program by filling in the blanks so that it correctly prints numbers from 1 to 10.

Program to print numbers from 1 to 10

for i in _____:

print(_____)

Also explain how the range() function works in this context.

Your Code

```
1 n=int(input("Enter the number:\n"))
2 for i in range(1,n+1):
3     print(i)
```

Input

10

Output

Enter the number:

1
2
3
4
5

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